

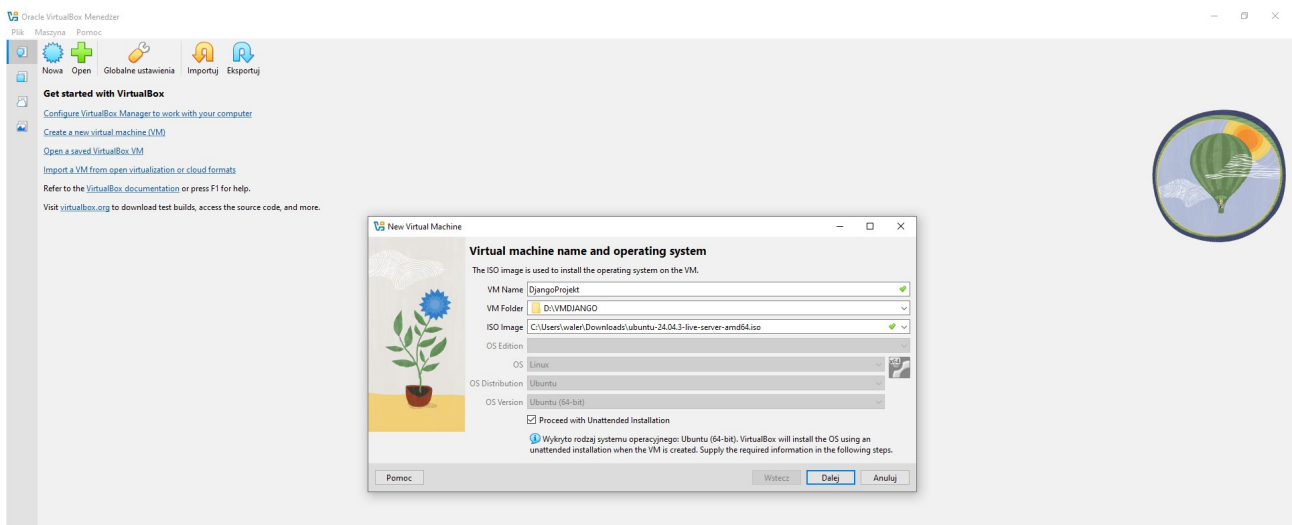
Tutorial dla grzecznych Bobasków z Samochodówki:

Temat: VPS na ubuntu + Vbox + nginx + django +gunicorn
„Moc w słabości się doskonali”
Coś dla nas....

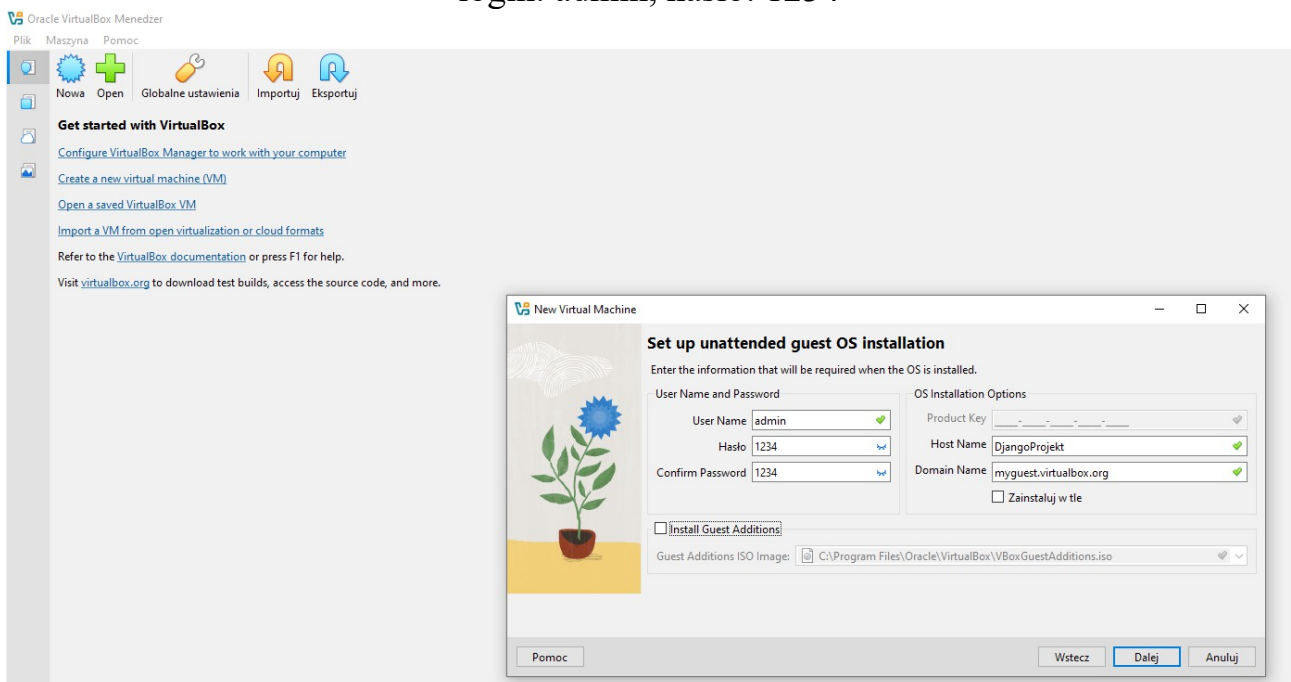
Pobieramy:

VirtualBox-7.2.6a-172322-Win.exe
ubuntu-24.04.3-live-server-amd64.iso

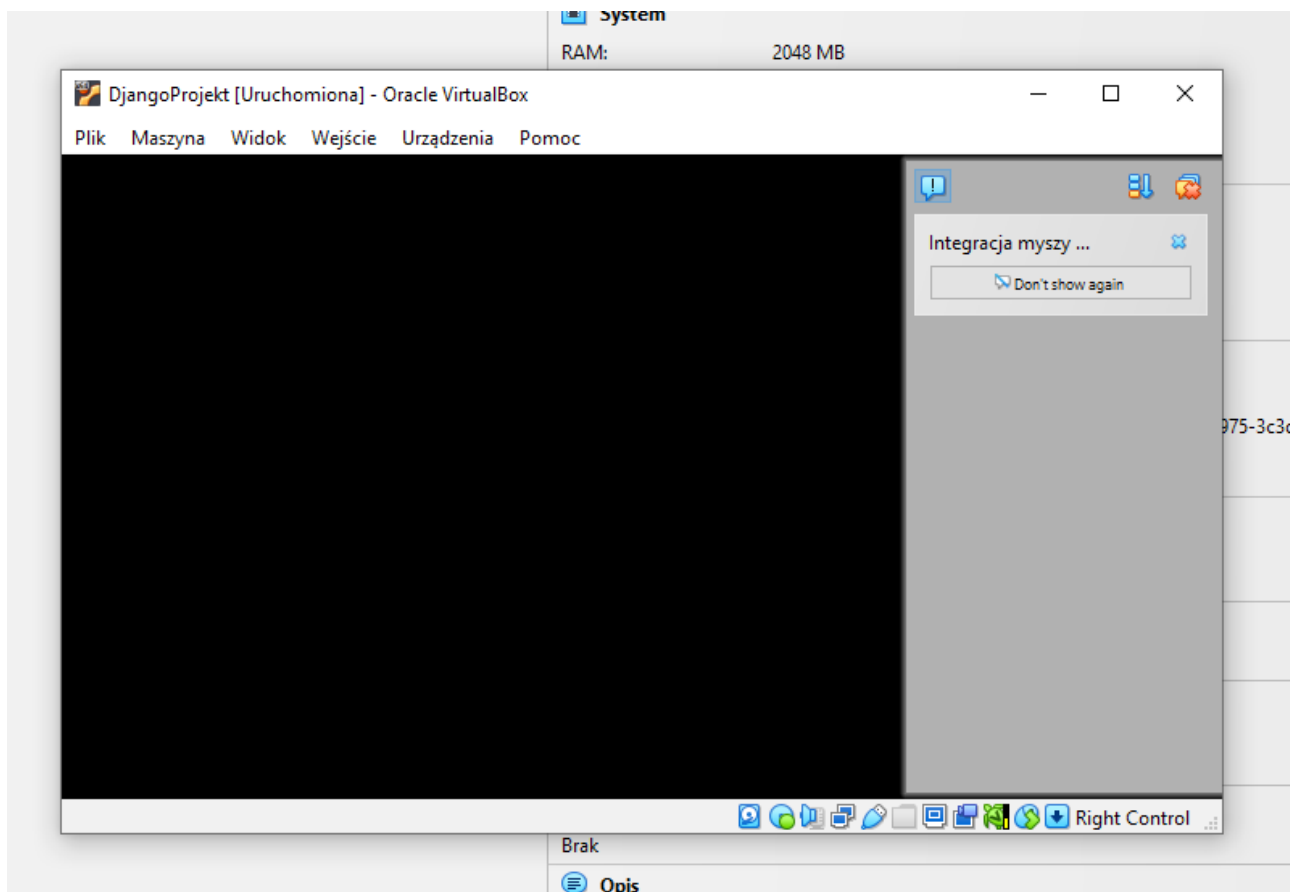
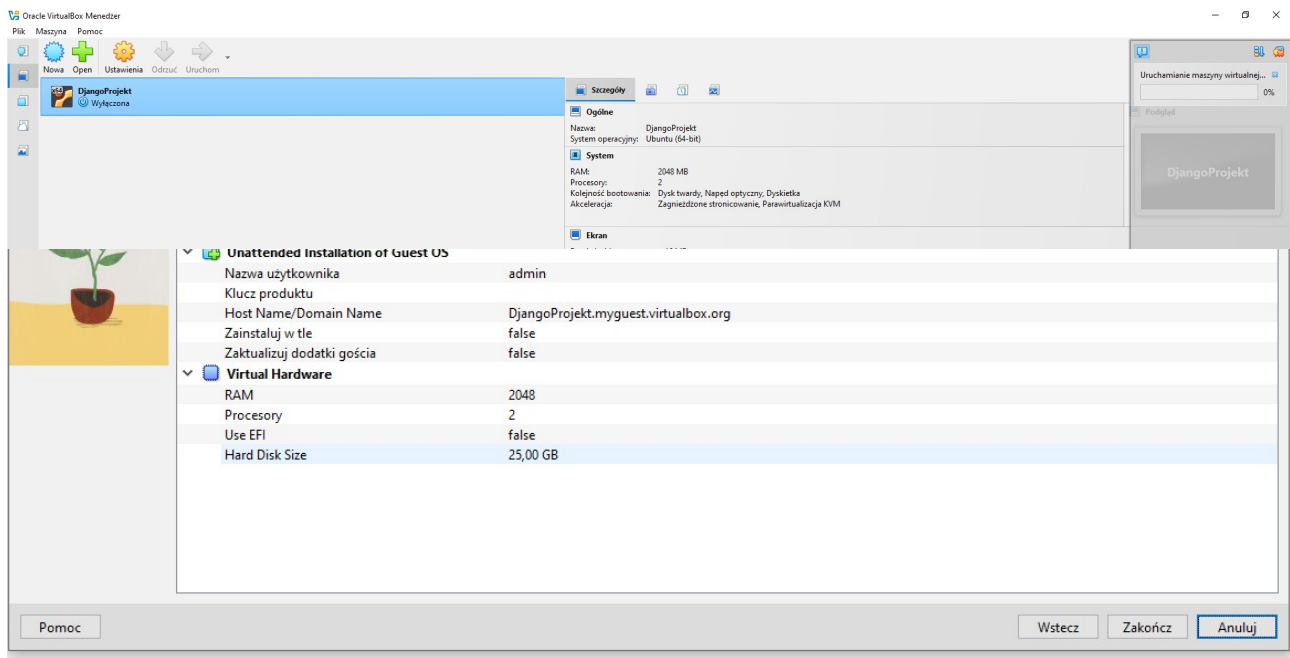
Tworzymy nową VM: „Create a new virtual machine VM”



Klikać „Dalej”.....
login: admin, hasło: 1234

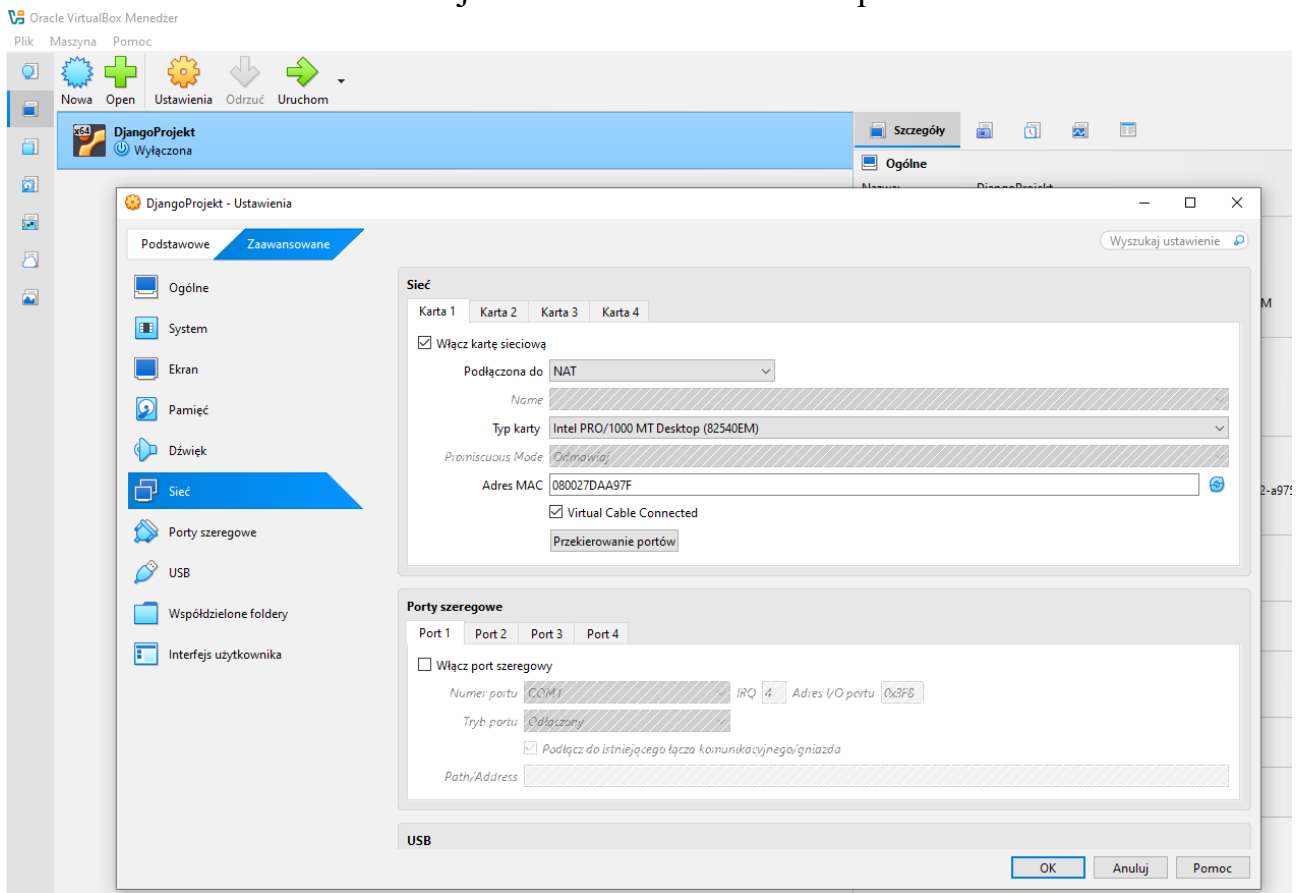


ustawiamy Virtual hardware

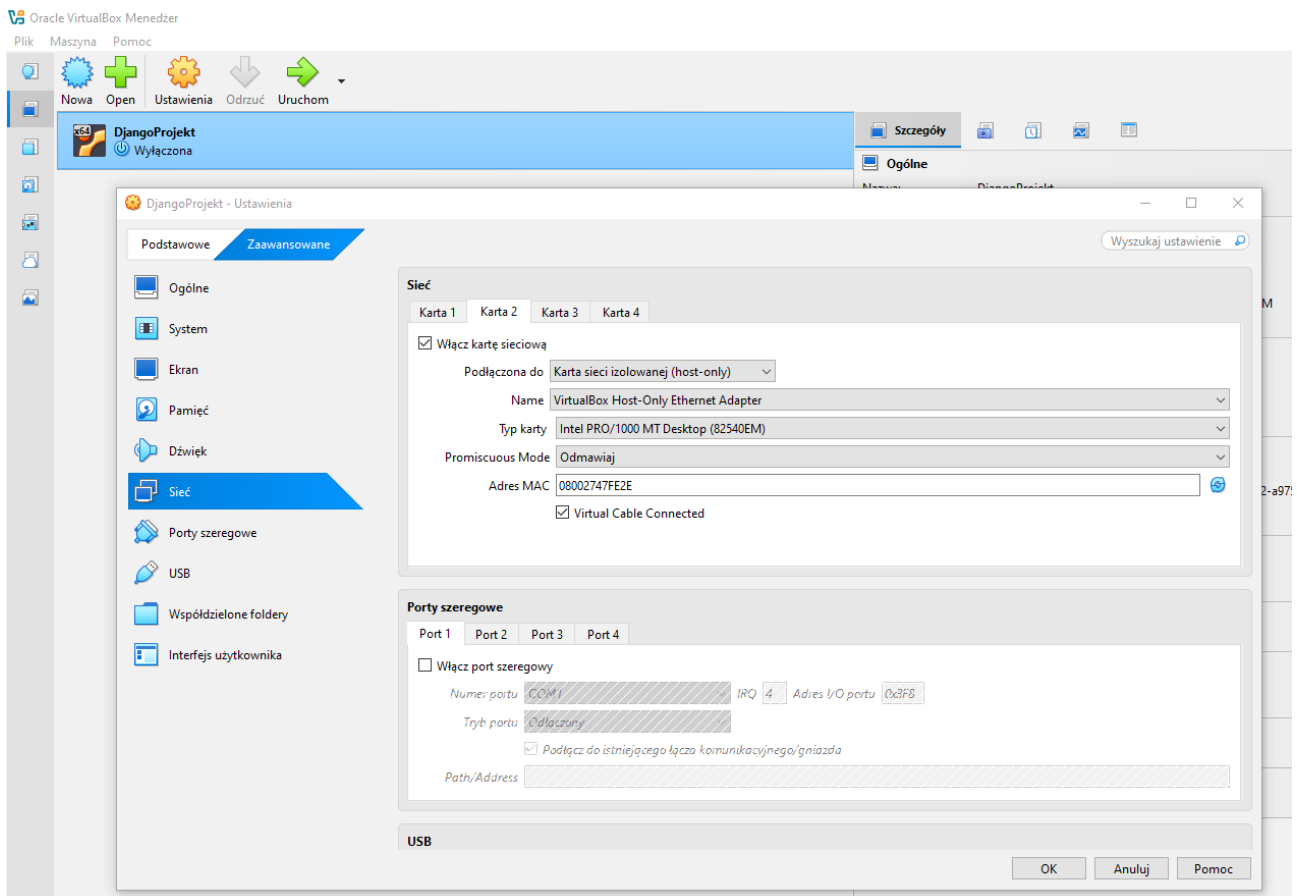


instalacja ubuntu.....przterywamy
i konfigurujemy interfejs sieciowy
2 karty sieciowe są potrzebne

1 karta jako Karta sieciowa z komputera



2 karta jako VirtualBox Host-Only



Uruchamiamy Vbox: instalacja ubuntu ...
po instalacji ubuntu logujemy się i instalujemy ssh
sudo apt install ssh, resztę robimy z putty ssh

Zanim będziemy się łączyć przez ssh sprawdzamy ip a

192.168.56.102

```
Cmder

C:\cmdr
λ ssh admin@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:wx0D/3DqkKa4dU1wiktctjntfGTrBAy5mmhFNqP0osj0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts.
admin@192.168.56.102's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Mon Feb  9 01:16:04 PM UTC 2026

System load:          0.0
Usage of /:           10.7% of 24.44GB
Memory usage:         11%
Swap usage:           0%
Processes:            113
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:feda:a97f

Expanded Security Maintenance for Applications is not enabled.

65 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

admin@DjangoProjekt:~$
```

To lecimy: robimy update i upgrade systemu do najnowszych aktualizacji

```
admin@DjangoProjekt:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for admin:
Hit:1 http://pl.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://pl.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://pl.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Fetched 126 kB in 1s (102 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
70 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

instalujemy pythona + pip + venv

Cmdr

```
admin@DjangoProjekt:~$ sudo apt install -y python3 python3-pip python3-venv
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential bzip2 cpp cpp-13 cpp-13-x86
  g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu gc
  libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan8 libatomic1 libbinutils libcc1-0 lib
```

Tworzymy nowy projekt

```
admin@DjangoProjekt:~$ mkdir django
```

Cmdr

```
admin@DjangoProjekt:~$ cd django/
admin@DjangoProjekt:~/django$ python3 -m venv venv
admin@DjangoProjekt:~/django$ ls
venv
admin@DjangoProjekt:~/django$ sudo source venv/bin/activate
sudo: source: command not found
admin@DjangoProjekt:~/django$ source venv/bin/activate
(venv) admin@DjangoProjekt:~/django$
```

Cmdr

```
(venv) admin@DjangoProjekt:~/django$ pip install django gunicorn
Collecting django
  Downloading django-6.0.2-py3-none-any.whl.metadata (3.9 kB)
Collecting gunicorn
  Downloading gunicorn-25.0.3-py3-none-any.whl.metadata (5.5 kB)
```

Cmdr

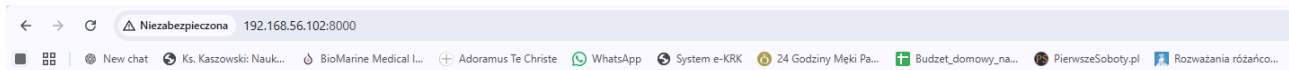
```
(venv) admin@DjangoProjekt:~/django$ django-admin startproject systempdf
(venv) admin@DjangoProjekt:~/django$ ls
systempdf  venv
(venv) admin@DjangoProjekt:~/django$ cd systempdf/
(venv) admin@DjangoProjekt:~/django/systempdf$ ls
manage.py  systempdf
(venv) admin@DjangoProjekt:~/django/systempdf$
```

Odblokujemy allow host i dodamy: 192.168.56.102

```
[sudo] password for admin:  
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano ./systempdf/settings.py
```

```
from pathlib import Path  
  
# Build paths inside the project like this: BASE_DIR / 'subdir'.  
BASE_DIR = Path(__file__).resolve().parent.parent  
  
# Quick-start development settings - unsuitable for production  
# See https://docs.djangoproject.com/en/6.0/howto/deployment/checklist/  
  
# SECURITY WARNING: keep the secret key used in production secret!  
SECRET_KEY = 'django-insecure-(f3*0y)c1smk^00sc#(ia=iu+3dbfxnh^_3z3i9@lio9zxky+x'  
  
# SECURITY WARNING: don't run with debug turned on in production!  
DEBUG = True  
  
ALLOWED_HOSTS = ['192.168.56.102']
```

```
(venv) admin@DjangoProjekt:~/django/systempdf$ python manage.py runserver 0.0.0.0:8000  
Watching for file changes with StatReloader  
Performing system checks...  
  
System check identified no issues (0 silenced).  
  
You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.  
Run 'python manage.py migrate' to apply them.  
February 09, 2026 - 13:55:06  
Django version 6.0.2, using settings 'systempdf.settings'  
Starting development server at http://0.0.0.0:8000/  
Quit the server with CONTROL-C.  
  
WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.  
For more information on production servers see: https://docs.djangoproject.com/en/6.0/howto/deployment/
```



The install worked successfully! Congratulations!

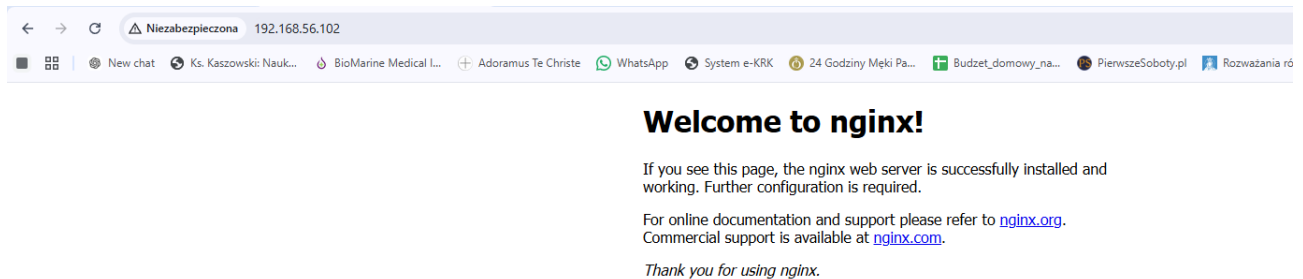
View [release notes](#) for Django 6.0

You are seeing this page because `DEBUG=True` is in your settings file and you have not configured any URLs.

django

```
(venv) admin@DjangoProjekt:~/django/systempdf$ gunicorn --bind 0.0.0.0:8000 mysite.wsgi
[2026-02-09 13:57:31 +0000] [11440] [INFO] Starting gunicorn 25.0.3
```

```
sudo apt install -y nginx
sudo systemctl enable nginx
sudo systemctl start nginx
```



Konfiguracja Nginx dla Django

```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano /etc/nginx/sites-available/arkadiusz
```

```
GNU nano 7.2
server {
    listen 80;
    server_name arkadiusz.local;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo ln -s /etc/nginx/sites-available/arkadiusz /etc/nginx/sites-enabled
(venv) admin@DjangoProjekt:~/django/systempdf$
```



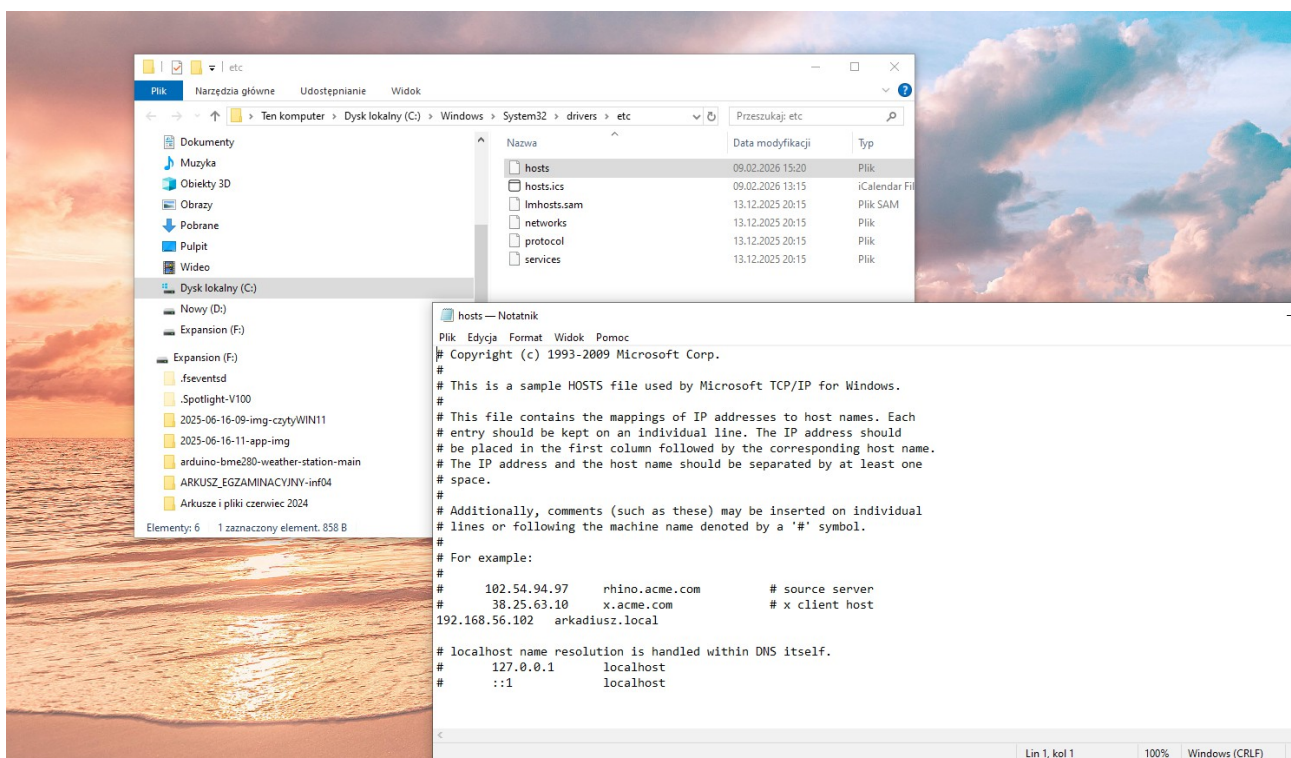
```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo systemctl reload nginx
(venv) admin@DjangoProjekt:~/django/systempdf$
```

```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano ./systempdf/settings.py
(venv) admin@DjangoProjekt:~/django/systempdf$
```

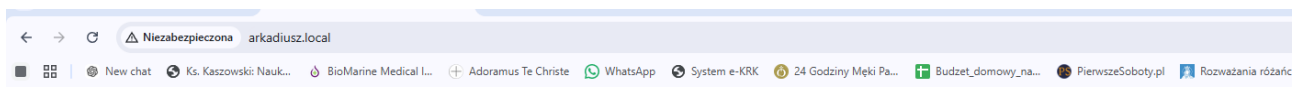
```
# SECURITY WARNING: don't run with debug turned on in production
DEBUG = True

ALLOWED_HOSTS = ['arkadiusz.local', '192.168.56.102']
```

```
(venv) admin@DjangoProjekt:~/django/systempdf$ gunicorn --bind 127.0.0.1:8000 systempdf.wsgi
[2026-02-09 14:18:04 +0000] [11993] [INFO] Starting gunicorn 25.0.3
[2026-02-09 14:18:04 +0000] [11993] [INFO] Listening at: http://127.0.0.1:8000 (11993)
[2026-02-09 14:18:04 +0000] [11993] [INFO] Using worker: sync
[2026-02-09 14:18:04 +0000] [11994] [INFO] Booting worker with pid: 11994
```



otwórz i dopisz:
C:\Windows\System32\drivers\etc\hosts
192.168.56.102 arkadiusz.local



The install worked successfully! Congratulations!

View [release notes](#) for Django 6.0

You are seeing this page because `DEBUG=True` is in your settings file and you have not configured any URLs.

django

Gunicorn jako systemd service

Socket Gunicorn

```
Cmdr
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano /etc/systemd/system/gunicorn.socket
```

```
Cmdr
GNU nano 7.2 /etc
[Unit]
Description=gunicorn socket

[Socket]
ListenStream=/run/gunicorn.sock

[Install]
WantedBy=sockets.target
```

Service Gunicorn

```
Cmder
admin@DjangoProjekt:~/django/systempdf$ sudo nano /etc/systemd/system/gunicorn.service
```

```
GNU nano 7.2
[Unit]
Description=gunicorn daemon
Requires=gunicorn.socket
After=network.target

[Service]
User=admin
Group=www-data
WorkingDirectory=/home/admin/django/systempdf
ExecStart=/home/admin/django/venv/bin/gunicorn \
    --access-logfile - \
    --workers 3 \
    --bind unix:/run/gunicorn.sock \
    systempdf.wsgi:application

[Install]
WantedBy=multi-user.target
```

Start + test

```
sudo systemctl daemon-reload
sudo systemctl start gunicorn.socket
sudo systemctl enable gunicorn.socket
```

```
Cmdr
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano /etc/systemd/system/gunicorn.service
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo systemctl daemon-reload
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo systemctl start gunicorn.socket
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo systemctl enable gunicorn.socket
Created symlink /etc/systemd/system/sockets.target.wants/gunicorn.socket → /etc/systemd/system/gunicorn.socket.
(venv) admin@DjangoProjekt:~/django/systempdf$ |
```

```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo systemctl status gunicorn
o gunicorn.service - gunicorn daemon
   Loaded: loaded (/etc/systemd/system/gunicorn.service; disabled; preset: enabled)
   Active: inactive (dead)
 TriggeredBy: ● gunicorn.socket
(venv) admin@DjangoProjekt:~/django/systempdf$
```

Nginx → socket (UPDATE)

```
TriggeredBy: ● gunicorn.socket
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano /etc/nginx/sites-available/arkadiusz
```

```
Cmdr
GNU nano 7.2 /et
server {
    listen 80;
    server_name arkadiusz.local;

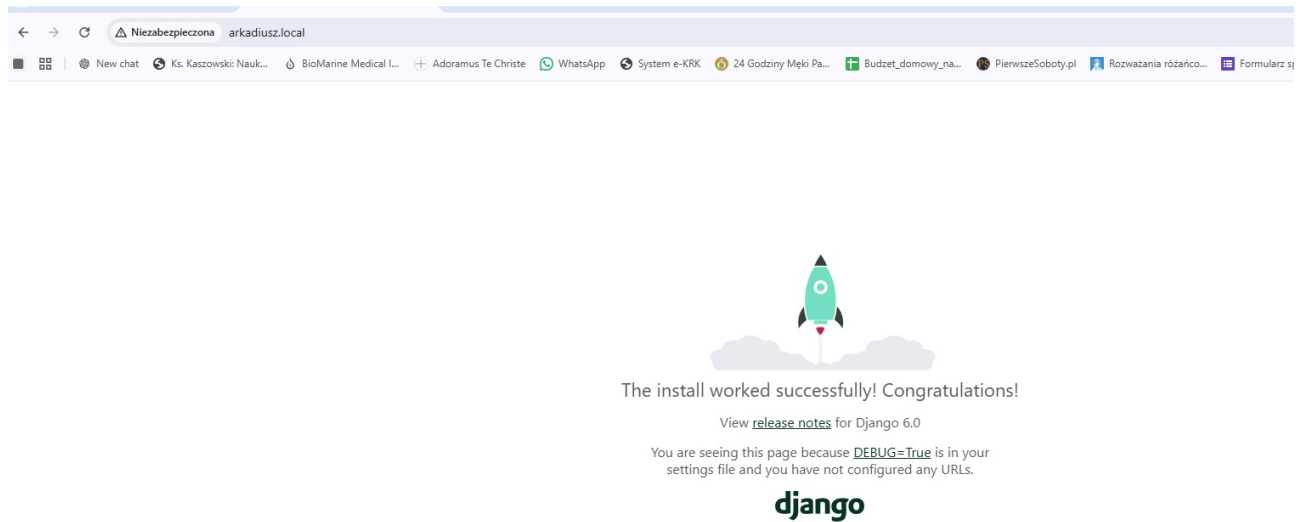
    location / {
        proxy_pass http://unix:/run/gunicorn.sock;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

Reload:

```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo systemctl reload nginx
(venv) admin@DjangoProjekt:~/django/systempdf$
```

reboot VM i test czy http://arkadiusz.local działa ???

działa yupiiii



Teraz aktywujemy pliki statyczne:
staticfiles + Nginx

```
Cmder

admin@DjangoProjekt:~$ ls
django
admin@DjangoProjekt:~$ cd django/
admin@DjangoProjekt:~/django$ ls
systempdf  venv
admin@DjangoProjekt:~/django$ source venv/bin/activate
(venv) admin@DjangoProjekt:~/django$ cd systempdf/
(venv) admin@DjangoProjekt:~/django/systempdf$ ls
db.sqlite3  manage.py  systempdf
(venv) admin@DjangoProjekt:~/django/systempdf$ |
```

```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano ./systempdf/settings.py
```

```
# Internationalization
# https://docs.djangoproject.com/en/6.0/topics/i18n/

LANGUAGE_CODE = 'en-us'

TIME_ZONE = 'UTC'

USE_I18N = True

USE_TZ = True


# Static files (CSS, JavaScript, Images)
# https://docs.djangoproject.com/en/6.0/howto/static-files/

STATIC_URL = 'static/'
STATIC_ROOT = BASE_DIR / 'staticfiles'
```

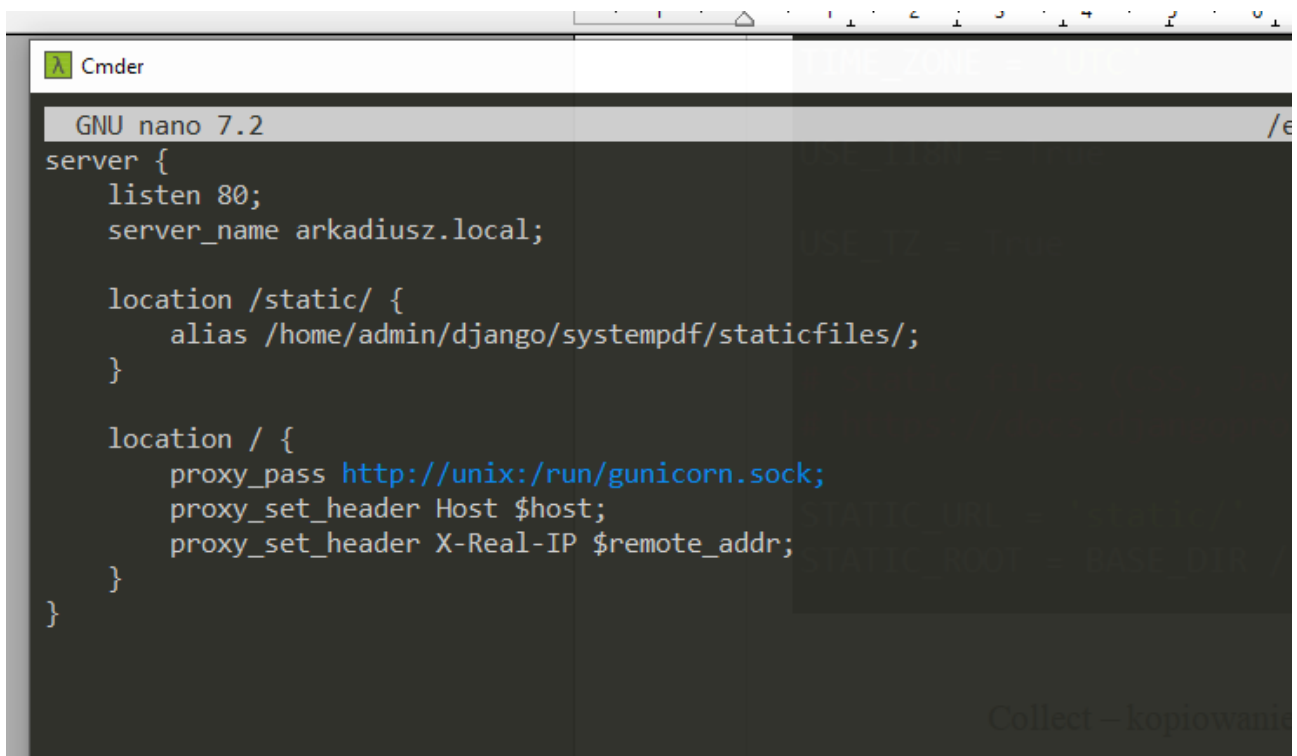
Collect – kopiowanie plików statycznych css itd.. do kat. staticfiles

```
(venv) admin@DjangoProjekt:~/django/systempdf$ python manage.py collectstatic
130 static files copied to '/home/admin/django/systempdf/staticfiles'.
(venv) admin@DjangoProjekt:~/django/systempdf$
```

Nginx static – konfiguracja static na serverze

sudo nano /etc/nginx/sites-available/arkadiusz

```
(venv) admin@DjangoProjekt:~/django/systempdf$ sudo nano /etc/nginx/sites-available/arkadiusz
(venv) admin@DjangoProjekt:~/django/systempdf$ |
```

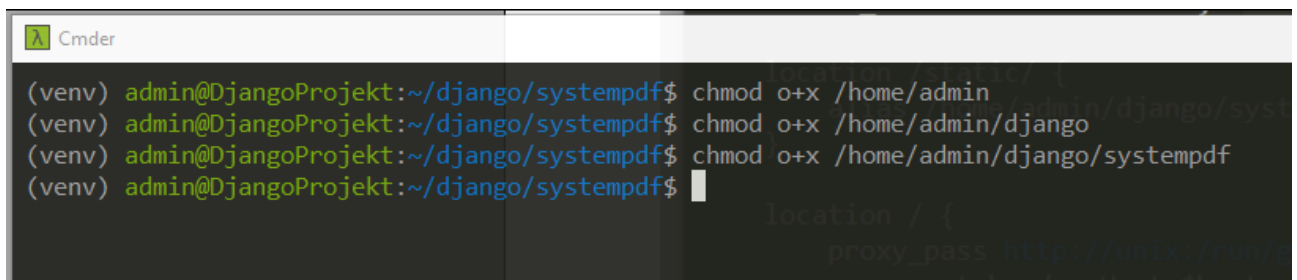


```
GNU nano 7.2 /etc/nginx/nginx.conf
server {
    listen 80;
    server_name arkadiusz.local;

    location /static/ {
        alias /home/admin/django/systempdf/staticfiles/;
    }

    location / {
        proxy_pass http://unix:/run/gunicorn.sock;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

dodajemy uprawnienia aby był dostęp do poszczególnych katalogów:



```
(venv) admin@DjangoProjekt:~/django/systempdf$ chmod o+x /home/admin
(venv) admin@DjangoProjekt:~/django/systempdf$ chmod o+x /home/admin/django
(venv) admin@DjangoProjekt:~/django/systempdf$ chmod o+x /home/admin/django/systempdf
(venv) admin@DjangoProjekt:~/django/systempdf$
```